Technical Note: Task-Visible Operator Mechanisms for In-Context Linear Ridge Regression

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1 Introduction

The goal is to test whether a trained transformer implements the linear-kernel ridge-regression operator, not just whether it predicts one label vector correctly. This distinction matters because many different algorithms can agree on one sampled label vector while responding differently to nearby label perturbations. We therefore turn each episode into an operator test: freeze the inputs and ask how predictions change as the context labels are varied.

Fix one episode and hold all non-label inputs fixed:

$$G = (X_{\text{ctx}}, X_{\text{tgt}}).$$

Let

$$X_{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times d_x}, \qquad X_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d_x}.$$

The transformer receives the fixed sequence

$$(x_1, y_1, 0), \dots, (x_{n_{\text{ctx}}}, y_{n_{\text{ctx}}}, 0), (x_1^\star, 0, 1), \dots, (x_{n_{\text{tgt}}}^\star, 0, 1),$$

where the first block contains context tokens and the second block contains target tokens. In the operator tests, only the context-label entries $y_1, \dots, y_{n_{\text{ctx}}}$ are perturbed. Rows of X_{ctx} and X_{tgt} are sampled from a Gaussian input law. Conditional on an episode covariance Σ,

$$x_i \sim \mathcal{N}(0, \Sigma), \qquad \Sigma = U \Lambda U^\top,$$

where U is a random orthogonal matrix obtained by QR-orthogonalizing a Gaussian matrix. During training, Λ is sampled from the spectral curriculum described in the experimental setup. The operator experiments reported below instead use the flat-spectrum evaluation geometry

$$\Sigma = cI, \qquad c \sim \text{Unif}[1, 10],$$

implemented by the shared flat eigenvalues sampler. Set

$$K := X_{\text{ctx}} X_{\text{ctx}}^\top, \qquad K_t := X_{\text{tgt}} X_{\text{ctx}}^\top, \qquad A := K + \sigma^2 I_{n_{\text{ctx}}}, \qquad \sigma^2 > 0.$$

These matrices are introduced because, once the inputs are fixed, all of ridge regression is encoded by the context Gram matrix K, the target-context Gram matrix K_t, and the regularized system matrix A. Labels are generated by a sampled teacher,

$$\beta \sim \mathcal{N}(0, I_{d_x}), \qquad y = X_{\text{ctx}} \beta + \eta, \qquad \eta \sim \mathcal{N}(0, \sigma^2 I_{n_{\text{ctx}}}).$$

The target values used for evaluation are the noiseless teacher values $X_{\mathrm{tgt}}\beta$. This is the same sampling convention used by the experiment scripts: context labels are noisy, target labels are not given to the transformer, and KRR is evaluated as the conditional predictor from context labels to target values. Ridge regression does not observe β. It forms its estimate from the sampled inputs and context labels. For context labels $y \in \mathbb{R}^{n_{\mathrm{ctx}}}$, exact linear-kernel ridge regression gives

$$\alpha^\star = A^{-1}y, \qquad y^\star = K_t\alpha^\star.$$

Equivalently, after the geometry is fixed, ridge regression is the conditional operator

$$T_G := K_tA^{-1}, \qquad y^\star = T_G y.$$

Equivalently, in primal form,

$$\widehat{\beta} = \left(X_{\mathrm{ctx}}^\top X_{\mathrm{ctx}} + \sigma^2 I_{d_x}\right)^{-1} X_{\mathrm{ctx}}^\top y, \qquad y^\star = X_{\mathrm{tgt}}\widehat{\beta}.$$

Every behavioral, representational, and causal test below is ultimately scored against this operator, not only against the realized prediction $T_G y$.

Let the trained transformer define the fixed-geometry label map

$$F_\theta^G : \mathbb{R}^{n_{\mathrm{ctx}}} \to \mathbb{R}^{n_{\mathrm{tgt}}}, \qquad F_\theta^G(y) = \widehat{y}_\theta.$$

When the geometry is clear, we write F and T for F_θ^G and T_G. The central comparison is therefore between two maps with the same input and output spaces: the transformer's local label-response map F and the exact KRR operator T.

A context-label direction is a vector $u \in \mathbb{R}^{n_{\mathrm{ctx}}}$ added to the context labels while the input geometry G is held fixed. Testing a direction means comparing the transformer's response to $y+\varepsilon u$ and $y-\varepsilon u$. Thus the word "direction" always refers to a direction in label space over the context positions, not to a direction in input-feature space or target space.

The transformer's activations are the hidden states produced by the trained transformer on the same fixed episode. When we speak about an activation subspace, we mean a subspace over context positions extracted from context-token hidden states. Later sections define how these subspaces are tested against the ridge operator.

This note separates the mechanism claim into three tests.

1. **Behavior.** Does the transformer's local response to label changes match T on held-out perturbations?

2. **Representation.** Do the transformer's activations contain a compact context-side subspace from which the same operator can be recovered?

3. **Causal use.** Once the relevant activation directions have been identified, does removing them damage the transformer's response?

The next section defines the rank requirement before inspecting the transformer. It is a rank result, not a learning guarantee. The experiments then check whether the trained transformer exposes a subspace meeting that requirement, whether the transformer response agrees with the operator built from that subspace, and whether the transformer causally relies on the extracted directions. The claim is local to task-distribution label directions; isotropic probes show that the transformer's learned response drifts off distribution.

All theoretical claims are for the linear kernel. The studied transformer is bidirectional, so the explanation is for the conditional prediction operator K_tA^{-1}, not for a target-independent stored dual solution.

High-level construction. The construction has one organizing idea. After the input geometry is fixed, ridge regression is a linear map from context labels to target predictions. We want to know whether the transformer realizes this map through a small context-side activation subspace. The plan is:

1. First define the ideal subspace problem without using the transformer: what is the smallest context-label subspace that can reproduce the ridge operator on task-distributed label directions?

2. Then extract candidate subspaces from the trained transformer's activations and test whether they are sufficient for the same ridge operator.

3. Finally test three things separately: whether the transformer response matches ridge regression, whether the extracted subspace is sufficient for ridge regression, and whether the transformer causally depends on the extracted directions.

Thus the main claim is: the transformer contains a low-dimensional activation subspace whose dimension matches the rank requirement defined below for the ridge operator; the operator built from that subspace is accurate; and removing the relevant extracted directions damages the transformer's task-local response.

Norms. The A-norm is the geometry induced by the ridge linear system. For $v \in \mathbb{R}^{n_{\text{ctx}}}$ and $M \in \mathbb{R}^{n_{\text{ctx}} \times d}$, define

$$\|v\|_A := \sqrt{v^\top A v}, \qquad \|M\|_{A,F} := \|A^{1/2} M\|_F.$$

All prediction-space norms are Euclidean or Frobenius norms. A fixed numerical floor $\eta_{\text{num}} > 0$ appears only in denominators; in experiments $\eta_{\text{num}} = 10^{-12}$.

What is not claimed. The pre-transformer theory supplies a rank requirement. It does not say that raw depth, head count, or width imply ridge behavior. It also does not give an SGD or training guarantee. The layerwise activation-span diagnostic is a post-training diagnostic of where the relevant directions appear, not a pre-training architecture budget.

2 Task-Visible Rank

This section gives the only pre-transformer theorem used in the paper. It depends only on the input distribution, the label law, and the desired accuracy. Its purpose is to define the target that any activation explanation must meet. Before looking inside the transformer, we ask: how many context-label directions are needed, in principle, to reproduce ridge predictions on the label directions the task actually produces? This number is the benchmark for the later activation construction. If the transformer exposes fewer useful directions than this, the proposed subspace cannot approximate the ridge operator on the task distribution; if it exposes enough, the remaining question is whether the transformer's local response is close to the corresponding subspace-based approximation and whether the transformer causally depends on that subspace.

Assume the matched random linear regression task

$$y = X_{\text{ctx}} \beta + \eta, \qquad \beta \sim \mathcal{N}(0, \tau^2 I_{d_x}), \qquad \eta \sim \mathcal{N}(0, \tau^2 \sigma^2 I_{n_{\text{ctx}}}).$$

The experiments use $\tau = 1$; the scale τ is included here only to make clear that an overall label scale does not affect the normalized operator errors below. Here the randomness in X_{ctx} and X_{tgt}

first samples the geometry G; the randomness in β and η then samples labels conditional on that geometry. Conditional on the geometry G,

$$\mathrm{Cov}(y \mid G) = \tau^2 A.$$

Thus task-label directions are isotropic in the whitened coordinates $A^{-1/2}y$. Define the task-visible operator

$$C_G := K_t A^{-1/2}.$$

Let $s_1(G) \geq s_2(G) \geq \cdots \geq 0$ be its singular values. The whitening is the reason C_G is the right object. In coordinates where task labels are standard Gaussian, C_G is exactly the ridge map from whitened label directions to target predictions. Its singular spectrum therefore says which task-label directions matter most for prediction.

Definition 1 (Task-visible rank)**.** *For fixed G and tolerance $\epsilon > 0$, define*

$$r_{\text{task}}(G;\epsilon) := \min\left\{ r : \frac{\sum_{i>r} s_i(G)^2}{\sum_i s_i(G)^2 + \eta_{\text{num}}} \leq \epsilon^2 \right\}.$$

For random geometries $G \sim \mathcal{D}_G$, define

$$r^{\text{hp}}_{\text{task}}(\epsilon,\delta) := \min\left\{ r : \mathbb{P}_G(r_{\text{task}}(G;\epsilon) \leq r) \geq 1-\delta \right\}.$$

This rank is the number of target-visible ridge directions needed on the task label distribution. It is a requirement that an activation subspace may or may not meet after training; it is not a raw architectural budget. For the linear kernel, $\mathrm{rank}(C_G) \leq \min\{\mathrm{rank}(X_{\text{ctx}}), n_{\text{tgt}}\} \leq \min\{d_x, n_{\text{tgt}}\}$, so the theorem below gives a concrete finite-dimensional requirement. On the flat-spectrum full-rank geometries used in the target-rank sweep with $d_x \leq 15$, the a priori rank calculation in the experiments verifies that this requirement is essentially d_x at the reported tolerance. For the additional $d_x = 30$ stress row, $n_{\text{tgt}} = 16$ caps the task-visible rank.

The next quantity turns the rank question into an operator approximation question. Suppose two candidate label-to-prediction maps, S_1 and S_2, are equal on one realized label vector but differ on other possible label directions. We need a way to say how different they are on the directions we care about. A probe law P specifies those directions. For example, the task probe law samples directions with the same covariance as task labels, while the isotropic law samples all context-label directions equally.

For a fixed geometry and a probe law P, define the population operator error

$$E_P(S_1,S_2) := \left(\frac{\mathbb{E}_{u\sim P}\|(S_1 - S_2)u\|_2^2}{\mathbb{E}_{u\sim P}\|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

The expectation is only over the probe direction u; the geometry is already fixed. The numerator is the mean squared prediction-response discrepancy between S_1 and S_2 on the chosen directions. The denominator is the mean squared response of the exact ridge operator on the same directions. Thus $E_P(S_1,S_2) = 0.01$ means that the root-mean-square operator discrepancy is about one percent of the root-mean-square ridge response under P. This normalization lets the theorem and the experiments use the same scale, and it prevents directions where ridge has almost no response from dominating the interpretation.

Theorem 1 (Task-visible rank theorem)**.** *Fix a geometry G in the matched linear task. Let $T = K_t A^{-1}$ and let $Q \in \mathbb{R}^{n_{\text{ctx}} \times r}$ be A-orthonormal, $Q^\top A Q = I_r$. Define the reduced operator*

$$T_Q := K_t Q Q^\top.$$

Let P_{task} be the task-label probe law $u \mid G \sim \mathcal{N}(0, A)$; the scalar factor τ^2 in the matched label covariance cancels in the normalized error. Then

$$\min_{Q^\top AQ = I_r} E_{P_{\text{task}}}(T_Q, T)^2 = \frac{\sum_{i>r} s_i(G)^2}{\sum_i s_i(G)^2 + \eta_{\text{num}}},$$

where $s_i(G)$ are the singular values of $C_G = K_t A^{-1/2}$. A minimizer is

$$Q_r^\star = A^{-1/2} V_{1:r},$$

where $V_{1:r}$ are the top right singular vectors of C_G. Consequently, if $r \geq r_{\text{task}}(G; \epsilon)$, then there exists an r-dimensional reduced operator with

$$E_{P_{\text{task}}}(T_Q, T) \leq \epsilon.$$

If $r \geq r_{\text{task}}^{\text{hp}}(\epsilon, \delta)$, the same statement holds with probability at least $1 - \delta$ over sampled geometries.

Proof. Let $W = A^{1/2} Q$. Then $W^\top W = I_r$ and

$$T_Q = K_t A^{-1/2} W W^\top A^{-1/2}.$$

Write $u = A^{1/2} z$. Under the task-label probe law, $z \sim \mathcal{N}(0, I)$. Therefore

$$Tu = C_G z, \qquad T_Q u = C_G W W^\top z,$$

so

$$(T - T_Q) u = C_G (I - W W^\top) z.$$

Taking expectation over z gives

$$\mathbb{E} \|(T - T_Q) u\|_2^2 = \|C_G (I - W W^\top)\|_F^2, \qquad \mathbb{E} \|Tu\|_2^2 = \|C_G\|_F^2.$$

The optimization is therefore

$$\min_{W^\top W = I_r} \|C_G (I - W W^\top)\|_F^2.$$

By Ky Fan's maximum principle, this residual is minimized by the top right singular vectors of C_G, and the minimum residual is $\sum_{i>r} s_i^2$. The high-probability statement follows from the definition of $r_{\text{task}}^{\text{hp}}$. □

Remark 1 (Meaning of the sufficient condition)**.** *The theorem above is a rank theorem for context-side subspaces. It says that an A-orthonormal trial space of sufficient task-visible dimension can represent the ridge operator on task-label probes. It does not say that a transformer learns, exposes, or uses such a space. Those are post-training questions.*

This leaves the central empirical problem. The theorem tells us what a sufficient context-side subspace would have to accomplish, but it does not tell us whether the trained transformer has such a subspace or whether the transformer's predictions depend on it. The next section therefore converts the rank condition into measurable diagnostics: first compare the transformer's local label-response map to ridge regression, then compare activation-derived reduced operators to both the transformer and the exact ridge operator.

3 Operator Diagnostics and Activation Certificates

Section 2 defined the ideal trial-space target. This section explains how to test that target in a trained transformer. The central principle is to freeze the geometry, perturb only labels, and compare the resulting local response to the operator T. We use local perturbations because the transformer is nonlinear as a function of the whole token sequence, even if the target algorithm is linear in the labels after the geometry is fixed.

Definition 2 (Central finite-difference response)**.** *For step size $\varepsilon > 0$ and direction $u \in \mathbb{R}^{n_{\mathrm{ctx}}}$, define*

$$D_\varepsilon F(y)[u] := \frac{F(y+\varepsilon u) - F(y-\varepsilon u)}{2\varepsilon}.$$

This is the measured label-to-prediction response of the transformer in direction u. If the transformer locally implements ridge regression, this quantity should be close to Tu on held-out label directions.

The population error from Section 2 is the ideal quantity, but experiments only use finitely many label directions. The next definition is its empirical version. The first error compares a nonlinear transformer to a fixed linear operator by asking: when the labels are nudged in direction u_j, does the transformer's measured response agree with what operator S would have produced on the same direction? Taking several held-out directions makes this an operator test rather than a test on the original label vector alone.

Definition 3 (Operator errors)**.** *Let $U = \{u_1, \ldots, u_m\}$ be a predeclared set of probe directions. For a fixed linear operator $S : \mathbb{R}^{n_{\mathrm{ctx}}} \to \mathbb{R}^{n_{\mathrm{tgt}}}$, define*

$$E_\varepsilon(F, S; U) := \left(\frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - Su_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\mathrm{num}}} \right)^{1/2}.$$

For two linear operators S_1, S_2, define

$$E(S_1, S_2; U) := \left(\frac{\sum_{j=1}^m \|(S_1 - S_2)u_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\mathrm{num}}} \right)^{1/2}.$$

The corresponding population quantity $E_P(S_1, S_2)$ was defined in Section 2.

Proposition 1 (Scale of the task-probe reduced-operator error)**.** *Let P_{task} be the task-label probe law $u = A^{1/2}z$, $z \sim \mathcal{N}(0, I)$, and let $Q^\top AQ = I$. Set*

$$C = K_t A^{-1/2}, \qquad W = A^{1/2}Q, \qquad P_Q = WW^\top.$$

Then

$$E_{P_{\mathrm{task}}}(T_Q, T)^2 = \frac{\|C(I - P_Q)\|_F^2}{\|C\|_F^2 + \eta_{\mathrm{num}}},$$

and therefore

$$0 \le E_{P_{\mathrm{task}}}(T_Q, T) \le 1.$$

If W is the span of the top r right singular vectors of C, then

$$E_{P_{\mathrm{task}}}(T_Q, T)^2 = \frac{\sum_{i>r} s_i^2}{\sum_i s_i^2 + \eta_{\mathrm{num}}},$$

where s_i are the singular values of C.

Proof. Since $W = A^{1/2}Q$, $W^\top W = Q^\top AQ = I$, so $P_Q = WW^\top$ is an orthogonal projection. For $u = A^{1/2}z$,

$$Tu = K_t A^{-1}A^{1/2}z = Cz,$$

while

$$T_Q u = K_t QQ^\top A^{1/2}z = K_t A^{-1/2}WW^\top z = CP_Q z.$$

Thus $(T - T_Q)u = C(I - P_Q)z$. Taking expectation over $z \sim \mathcal{N}(0, I)$ gives

$$\mathbb{E}\|(T - T_Q)u\|_2^2 = \|C(I - P_Q)\|_F^2, \qquad \mathbb{E}\|Tu\|_2^2 = \|C\|_F^2.$$

Because right multiplication by an orthogonal projection is a contraction in Frobenius norm, $\|C(I - P_Q)\|_F \leq \|C\|_F$, proving the bound. The singular-value formula follows from the Eckart–Young–Mirsky theorem, or equivalently Ky Fan's maximum principle, applied to the best rank-r right singular subspace of C. □

Proposition 2 (No universal upper bound for transformer-response errors)**.** *Fix a finite probe set U with ridge-response energy*

$$R_U^2 = \sum_{j=1}^{m} \|Tu_j\|_2^2 > 0.$$

If the measured local response is zero on all probes, $D_\varepsilon F(y)[u_j] = 0$, then

$$E_\varepsilon(F, T; U) = \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 1.$$

If the measured response is the negative ridge response, $D_\varepsilon F(y)[u_j] = -Tu_j$, then

$$E_\varepsilon(F, T; U) = \frac{2R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 2.$$

In contrast, $E_\varepsilon(F, T; U)$ has no universal upper bound.

Proof. The first two identities follow by substituting $D_\varepsilon F(y)[u_j] = 0$ and $D_\varepsilon F(y)[u_j] = -Tu_j$ into the definition. For unboundedness, suppose on the probe set the measured response is $D_\varepsilon F(y)[u_j] = cTu_j$. Then

$$E_\varepsilon(F, T; U) = |c - 1| \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}},$$

which diverges as $|c| \to \infty$ whenever $R_U > 0$. □

Proposition 3 (Triangle decomposition)**.** *For any fixed linear operator S and any finite probe set U,*

$$E_\varepsilon(F, T; U) \leq E_\varepsilon(F, S; U) + E(S, T; U).$$

In particular, taking $S = T_Q$ separates transformer-to-subspace error from subspace-to-ridge error.

Proof. Stack the probe responses as columns. With

$$M_F = [D_\varepsilon F(y)[u_1] \ \cdots \ D_\varepsilon F(y)[u_m]], \qquad M_S = [Su_1 \ \cdots \ Su_m], \qquad M_T = [Tu_1 \ \cdots \ Tu_m],$$

the numerator of each error is a Frobenius norm, and all three errors have the same denominator $(\|M_T\|_F^2 + \eta_{\text{num}})^{1/2}$. The ordinary triangle inequality gives

$$\|M_F - M_T\|_F \leq \|M_F - M_S\|_F + \|M_S - M_T\|_F,$$

and division by the common denominator proves the claim. □

Remark 2 (How to read the error values)**.** *For the population task-probe reduced-operator error, the scale is bounded between 0 and 1. An error e is a relative RMS response error, and* e^2 *is the fraction of task-probe ridge-response energy left unexplained by the reduced operator. Thus* $e = 0.01$ *means about one percent RMS error,* $e = 0.1$ *means about one percent squared response energy is missed, and* $e = 0.54$ *means roughly* $0.54^2 \approx 29\%$ *of the task response energy is missed. Values near 1 mean that the reduced operator leaves almost all task-visible ridge response unexplained.*

For transformer errors $E_\varepsilon(F, T)$ *and* $E_\varepsilon(F, T_Q)$*, there is no analogous upper bound. A value near 1 is already comparable to a zero-local-response baseline, values above 1 are worse than having no ridge-like label response on the probes, and very large values indicate that the finite-difference response is badly misaligned or unstable relative to the ridge response. All these readings are probe-law dependent: task-probe and isotropic-probe errors answer different questions.*

The first quantity, $E_\varepsilon(F, S; U)$, is used whenever one side of the comparison is the trained transformer. If $S = T$, it measures whether the transformer behaves like exact ridge regression on the probes. If $S = T_Q$, it measures whether the transformer's local response is close to the reduced operator induced by an activation subspace. The denominator is the total ridge-response energy on the same probe set, so the score is read as a relative root-mean-square response error. A small value means the discrepancy is small compared with the prediction changes that ridge regression itself would make on those directions.

The second quantity, $E(S_1, S_2; U)$, removes the transformer from the comparison and only asks whether two linear operators act similarly on the same probes. This is why later sections can separate transformer-to-reduced-operator error, $E_\varepsilon(F, T_Q)$, from subspace error, $E(T_Q, T)$.

Task-distribution probes test the directions emphasized by training; isotropic probes test a stronger global response. If $u \sim \mathcal{N}(0, I)$, then $\mathbb{E}\|Mu\|_2^2 = \|M\|_F^2$, so isotropic Gaussian probes estimate Frobenius operator error.

The operator errors above assume that the transformer has a stable local linear response around the observed label vector. A small transformer-to-ridge error is hard to interpret if the finite-difference response changes strongly when two probe directions are combined. The additivity defect is a guardrail for this issue: it checks whether the local response to $u + v$ is approximately the sum of the local responses to u and v.

Definition 4 (Additivity defect)**.** *For paired probe directions* $V = \{(u_j, v_j)\}_{j=1}^m$*, define*

$$C_j := F(y + \varepsilon u_j + \varepsilon v_j) - F(y + \varepsilon u_j - \varepsilon v_j) - F(y - \varepsilon u_j + \varepsilon v_j) + F(y - \varepsilon u_j - \varepsilon v_j),$$

$$A_\varepsilon(F; V) := \left(\frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j + \varepsilon v_j)\|_2^2 + \eta_{\mathrm{num}}} \right)^{1/2}.$$

The four evaluations in C_j cancel exactly for an affine function of the labels. Thus a small $A_\varepsilon(F; V)$ says that, at the chosen scale and probe law, the transformer's local response is close enough to linear that comparing it to a linear operator is meaningful. A large value would mean that the transformer may be nonlinear in the neighborhood being probed, so errors such as $E_\varepsilon(F, T)$ would mix two effects: failure to be ridge and failure to be locally linear. The additivity defect is only a validity check for the operator diagnostic; by itself it is not evidence that the linear operator is the ridge operator.

3.1 Reduced Operators from Activations

The behavioral test says whether the transformer response matches ridge regression, but not where the computation is represented. The next step is to ask whether the context-token activations define a small trial space from which ridge regression can be reconstructed. This is why the basis Q lives on the context-label side rather than in target space.

Definition 5 (A-orthonormal trial basis and reduced operator)**.** *A matrix $Q \in \mathbb{R}^{n_{\text{ctx}} \times r}$ is A-orthonormal if $Q^\top A Q = I_r$. The reduced operator induced by Q is the linear operator*

$$T_Q := K_t Q Q^\top.$$

For a general full-rank basis $Z \in \mathbb{R}^{n_{\text{ctx}} \times r}$, the same operator is

$$T_Z := K_t Z (Z^\top A Z)^{-1} Z^\top.$$

The reduced operator is the restriction of ridge regression to the trial space. If $Q^\top A Q = I$, then the reduced dual vector

$$\alpha_Q = Q Q^\top y$$

is the A-orthogonal projection of $\alpha^\star = A^{-1} y$ onto $\text{col}(Q)$, and

$$T - T_Q = K_t (I - Q Q^\top A) A^{-1}.$$

Thus T_Q is not an arbitrary readout from activations. It is the ridge operator one obtains when the dual solution is forced to lie in the activation-derived context subspace.

Proposition 4 (Linear feature trial space)**.** *For the linear kernel, let Z be any full-rank basis for* $\text{col}(X_{\text{ctx}})$. *Then the reduced operator*

$$T_Z = K_t Z (Z^\top A Z)^{-1} Z^\top$$

equals the exact operator $T = K_t A^{-1}$.

Proof. Fix y and write $\alpha^\star = A^{-1} y$ and

$$\alpha_Z = Z (Z^\top A Z)^{-1} Z^\top y.$$

The restricted normal equations give

$$Z^\top A (\alpha^\star - \alpha_Z) = 0.$$

Since $\text{col}(Z) = \text{col}(X_{\text{ctx}})$, this implies

$$X_{\text{ctx}}^\top A (\alpha^\star - \alpha_Z) = 0.$$

Let $r = \alpha^\star - \alpha_Z$. Using $A = X_{\text{ctx}} X_{\text{ctx}}^\top + \sigma^2 I$,

$$X_{\text{ctx}}^\top A r = (X_{\text{ctx}}^\top X_{\text{ctx}} + \sigma^2 I_{d_x}) X_{\text{ctx}}^\top r.$$

The matrix in parentheses is positive definite, so $X_{\text{ctx}}^\top r = 0$. Therefore $K_t r = X_{\text{tgt}} X_{\text{ctx}}^\top r = 0$, and $T_Z y = K_t \alpha_Z = K_t \alpha^\star = T y$ for every y. □

Definition 6 (Frozen extraction map)**.** *A frozen extraction map $\mathcal{E}$ maps the unperturbed episode and activations to an A-orthonormal basis*

$$Q = \mathcal{E}(G, y, \text{activations}).$$

After Q is extracted, the reduced operator T_Q is held fixed when label perturbations are tested.

Freezing Q is essential. Otherwise the basis could change with every probe direction, and the test would no longer certify one stable reduced operator carried by the unperturbed activations. In the experiments, this abstract map is instantiated in three concrete ways. Experiment 1 uses final-state weighted-SVD extractors: a raw extractor from the unperturbed final context hidden state, and a response-only extractor from final-state finite differences on build probes. Experiment 2 uses the layerwise activation-span extractor, which builds a basis from hidden-state finite-difference responses after removing directions already explained by earlier layers. Experiment 3 uses the raw final-state extractor from Experiment 1, because the causal intervention removes directions from the actual residual stream. In all cases the extraction uses only the unperturbed episode and build probes; evaluation is performed afterward on held-out probes with Q fixed.

Definition 7 (Pointwise errors)**.** *For a frozen basis Q, define*

$$e_{\mathrm{sub}}(Q;G,y) = \frac{\|T_Q y - Ty\|_2}{\|Ty\|_2 + \eta_{\mathrm{num}}}, \qquad \tau_{\mathrm{read}}(Q;G,y) = \frac{\|F(y) - T_Q y\|_2}{\|Ty\|_2 + \eta_{\mathrm{num}}}.$$

Then

$$\frac{\|F(y) - Ty\|_2}{\|Ty\|_2 + \eta_{\mathrm{num}}} \le e_{\mathrm{sub}}(Q;G,y) + \tau_{\mathrm{read}}(Q;G,y).$$

This pointwise statement is only a sanity check. The operator statement is stronger.

Theorem 2 (Operator residual decomposition)**.** *For every frozen A-orthonormal basis Q and probe set U,*

$$D_\varepsilon F(y)[u] - Tu = \underbrace{D_\varepsilon F(y)[u] - T_Q u}_{\textit{transformer–reduced-operator residual}} + \underbrace{(T_Q - T)u}_{\textit{subspace residual}}.$$

Consequently,

$$E_\varepsilon(F,T;U) \le E_\varepsilon(F,T_Q;U) + E(T_Q,T;U).$$

The two terms answer different questions. The subspace residual $E(T_Q,T)$ asks whether the activation subspace is sufficient for the ridge operator. The transformer–reduced-operator residual $E_\varepsilon(F,T_Q)$ asks whether the trained transformer's local response is close to the reduced operator induced by that subspace. These two residuals are the quantities used for the mechanism claim; auxiliary explained-fraction summaries are computed in the experiment logs but are not needed for the note's argument.

3.2 Final-State Extractors

The final-state extractor is the simplest way to ask whether the computation has left a usable context-side subspace in the residual stream. The object we need for the reduced operator is a basis $Q \in \mathbb{R}^{n_{\mathrm{ctx}} \times r}$: its columns are directions over context positions. The final context hidden state has exactly this context axis. If $H_L^{\mathrm{ctx}} \in \mathbb{R}^{n_{\mathrm{ctx}} \times D}$ is the final context-token hidden-state matrix, then each hidden coordinate gives one length-n_{ctx} pattern across the context tokens. The span of these patterns is the part of context-label space that is linearly visible in the final context representation.

The extractor measures this visible span and converts it into the kind of basis used by the reduced operator. For any matrix $M \in \mathbb{R}^{n_{\mathrm{ctx}} \times p}$ whose columns are context-token patterns, compute

$$A^{1/2} M = U \Sigma V^\top.$$

We keep the left singular directions whose singular values are at least $\tau_{\mathrm{sv}} \sigma_1$, up to the cap $r_{\max}$, and set

$$Q(M) := A^{-1/2} U_{\mathrm{kept}}.$$

This guarantees $Q(M)^\top A Q(M) = I$. The $A^{1/2}$ weighting is not a cosmetic choice: the ridge system measures dual directions using A, so the extractor ranks activation patterns by size in the same geometry in which $T_Q = K_t Q Q^\top$ is defined.

Experiment 1 uses two choices of M. The response-only choice is built from task-label build directions and then held fixed when we test both task-label and isotropic evaluation probes.

The raw extractor uses

$$M_{\text{raw}} = H_L^{\text{ctx}}, \qquad Q_L^{\text{raw}} := Q(M_{\text{raw}}).$$

This asks a broad availability question: does the final context state contain some subspace from which the ridge operator can be reconstructed?

The response-only extractor uses hidden-state changes caused by label perturbations. For build probes $u \in U_{\text{build}}$, define

$$Z_L(u) = \frac{H_L^{\text{ctx}}(y + \varepsilon u) - H_L^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon},$$

and stack these matrices as

$$M_{\text{resp}} = [Z_L(u_1), \ldots, Z_L(u_m)], \qquad Q_L^{\text{resp}} := Q(M_{\text{resp}}).$$

This asks a stricter question: do the label-sensitive final-state directions contain a subspace whose reduced operator is close to ridge regression?

After either basis is extracted, it is frozen. The experiment then computes the reduced operator $T_Q = K_t Q Q^\top$ and measures two held-out quantities: $E(T_Q, T)$, which tests whether the extracted subspace is sufficient for ridge, and $E_\varepsilon(F, T_Q)$, which tests whether the transformer's local response is close to the reduced operator induced by that subspace.

For a bidirectional transformer, these bases may be target-conditioned. The claim is therefore about the conditional prediction operator $K_t A^{-1}$.

The final-state certificate is enough to test whether a suitable subspace is present at the end of the computation. It is not enough to say how that subspace arises across layers. The next section introduces a stricter diagnostic for that question.

4 Layerwise Activation Span

The final-state certificate says whether a useful subspace is present at the end of the computation. It does not say whether that subspace can be traced to label-sensitive changes that appear during the layers. This section builds a layerwise activation span for that purpose. The diagnostic is deliberately post-training: it measures what the trained transformer exposes, not what depth, width, or head count should guarantee.

The construction keeps only directions that move when the context labels are perturbed. Fix build probes $U_{\text{build}} = \{u_1, \ldots, u_m\}$. Let $H_\ell^{\text{ctx}}(y)$ be the context-token hidden-state matrix after layer ℓ, and define the finite-difference hidden response

$$Z_\ell(u) := \frac{H_\ell^{\text{ctx}}(y + \varepsilon u) - H_\ell^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stack the build-probe responses as

$$M_\ell = [Z_\ell(u_1), \ldots, Z_\ell(u_m)].$$

The columns of M_ℓ are context-position patterns that are visible in layer ℓ when labels move. A direction that does not appear in these finite differences is not counted as evidence for a label-computation mechanism.

We now accept new directions one layer at a time. If Q is an A-orthonormal basis of a subspace, the A-orthogonal projection of a matrix M onto that subspace is

$$P_Q^A M := QQ^\top AM.$$

Before processing layer ℓ, suppose that the accepted direction matrices $S_0, \ldots, S_{\ell-1}$ have already been constructed. Each S_j has A-orthonormal columns and represents directions first accepted at layer j. The directions already available before layer ℓ are

$$R_\ell^{\mathrm{pre}} := \mathrm{span}\{A^t S_j : 0 \leq j < \ell,\ 0 \leq t \leq \ell - j\}.$$

For $\ell = 0$ this space is empty. The powers of A are an accounting device: a direction accepted earlier has more remaining layers in which it could be refined in the ridge-system geometry. We do not need to claim that the network literally multiplies by A at each layer; the point is to avoid counting as new a direction that is already obtainable from earlier accepted directions under the same geometry used by the reduced operator.

Let Q_ℓ^{pre} be an A-orthonormal basis of R_ℓ^{pre}. Remove the part of the layer response that is already explained by this earlier space:

$$N_\ell := M_\ell - P^A_{Q_\ell^{\mathrm{pre}}} M_\ell = \left(I - Q_\ell^{\mathrm{pre}} (Q_\ell^{\mathrm{pre}})^\top A\right) M_\ell.$$

Thus N_ℓ is the novel label-sensitive part of layer ℓ.

Candidate directions for layer ℓ are obtained by an A-weighted SVD of N_ℓ. This means computing

$$A^{1/2} N_\ell = U_\ell \Sigma_\ell V_\ell^\top.$$

The left singular vectors with $\sigma_i \geq \tau_{\mathrm{sv}} \sigma_1$ are the candidates. A declared selection rule then chooses an index set I_ℓ of these candidates, with $s_\ell = |I_\ell|$. The experiment below uses a causal selection rule: a candidate is accepted only if removing it from the context residual stream changes the transformer's local response by a predeclared amount. If the accepted indices are I_ℓ, define

$$S_\ell := A^{-1/2} U_{\ell, I_\ell}.$$

Then $S_\ell^\top A S_\ell = I$. The A-weighting matters because the reduced operator and the ridge dual variables are measured in the A-geometry. This SVD is therefore principal-component extraction in the geometry relevant to the ridge system, not in the raw Euclidean geometry of hidden states.

After the last layer L, define the layerwise activation span

$$R_{\mathrm{lay}} := \mathrm{span}\{A^t S_\ell : 0 \leq \ell \leq L,\ 0 \leq t \leq L - \ell\}, \qquad d_{\mathrm{lay}} := \dim R_{\mathrm{lay}}.$$

Let Q_{lay} be an A-orthonormal basis of R_{lay}. The reduced operator induced by the layerwise span is

$$T_{Q_{\mathrm{lay}}} = K_t Q_{\mathrm{lay}} Q_{\mathrm{lay}}^\top.$$

We also report the upper count

$$B_{\mathrm{lay}} := \sum_{\ell=0}^{L} s_\ell (L - \ell + 1), \qquad d_{\mathrm{lay}} \leq B_{\mathrm{lay}}.$$

This count is not an architecture budget. It is only the number of candidate directions produced by the accounting procedure before linear dependencies are removed.

The span is useful only if it actually covers the layer responses it is meant to explain. For each layer k, define the partial span

$$R_k := \operatorname{span}\{A^t S_\ell : 0 \le \ell \le k,\ 0 \le t \le k-\ell\},$$

and let Q_k be an A-orthonormal basis of R_k. The closure defect is

$$\delta_k^{\text{close}} := \frac{\|(I - Q_k Q_k^\top A) M_k\|_{A,F}}{\|M_k\|_{A,F} + \eta_{\text{num}}}.$$

A small closure defect means that the accepted directions up to layer k can reconstruct the label-sensitive response observed at layer k. A large closure defect means that the final span may still work as an endpoint certificate, but the proposed layerwise account is missing much of the hidden response it claims to summarize.

The refinement fraction is an auxiliary diagnostic for how much a layer reuses directions accepted earlier. For $k \ge 1$, let

$$B_k := \operatorname{span}(R_{k-1} \cup A R_{k-1}),$$

and let Q_k^{ref} be an A-orthonormal basis of B_k. Define

$$\rho_k^A := \frac{\|Q_k^{\text{ref}} (Q_k^{\text{ref}})^\top A M_k\|_{A,F}}{\|M_k\|_{A,F} + \eta_{\text{num}}}, \qquad \rho_0^A := 0.$$

Large ρ_k^A means that much of layer k's label response is already explained by previous directions plus one additional A-refinement. Small ρ_k^A means that the layer response is mostly treated as new information. The main operator claim does not rely on this fraction; it is reported to separate reuse of earlier streams from genuinely new layerwise directions.

Experiment 2 tests the following post-training hypothesis:

$$d_{\text{lay}}/r_{\text{task}} \approx 1 \quad \Longrightarrow \quad E_{P_{\text{task}}}(T_{Q_{\text{lay}}}, T) \approx 0,$$

provided the closure defects are small. Agreement between the transformer response and the layerwise span is tested separately by $E_\varepsilon(F, T_{Q_{\text{lay}}})$.

The high-level idea is now clear. The task-visible rank theorem says that ridge regression does not need all possible context-label directions: under task-label probes, it can be represented by a low-dimensional context-side space. The layerwise diagnostic asks whether the transformer appears to build such a space in its context-token activations. If it does, then label perturbations should reveal only a small number of directions, most later responses should be refinements or reuse of earlier directions, and the final span should induce a reduced operator close to exact ridge regression. The desired evidence is therefore simultaneous: d_{lay} should be close to r_{task}, $T_{Q_{\text{lay}}}$ should have small operator error, and the closure defects should be small. This combination would mean that the activation span is not merely a post hoc low-rank fit at the output; it is also consistent with the transformer carrying the task-relevant context-label degrees of freedom through its layers.

At this point we have a behavioral test, an endpoint subspace certificate, and a layerwise activation-span diagnostic. The experiments now apply these objects in order. Experiment 1 tests behavior and the final-state certificate, including its low-rank alternative. Experiment 2 tests the layerwise span against the task-visible rank. Experiment 3 tests whether the directions that the reduced operator ranks as important are causally used.

5 Experiments

The experiments follow the certificate in order. The common design principle is to separate construction from evaluation: subspaces and fitted controls are built from one probe set, while numerical claims are reported on held-out label directions. This is the experimental analogue of freezing Q in the theory. The experiments are therefore not three independent demonstrations. They are a sequence of checks that close the gaps left by the theorem: behavior first, representation second, and causal use third.

5.1 Experimental Setup

Transformer checkpoint. Unless otherwise stated, we use checkpoint `model_L8.pt`: an 8-layer bidirectional transformer with $d_x = 5$, width $D = 128$, 4 attention heads, two-layer GELU FFNs, residual connections, no LayerNorm, no positional encodings, and a linear readout on target tokens.

Training distribution. Each training episode samples a fresh linear regression task. The input covariance is $\Sigma = U\Lambda U^\top$, with U drawn by QR-orthogonalizing a Gaussian matrix and Λ drawn uniformly from three spectral families: polynomial decay, exponential decay, and step spectra. Labels are $y = x^\top\beta + \epsilon$ with $\beta \sim \mathcal{N}(0, I)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2)$, using $\sigma^2 = 0.1$. Training uses 50 points per episode with $n_{\text{tgt}} \sim \text{Unif}\{1, \dots, 10\}$ and $n_{\text{ctx}} = 50 - n_{\text{tgt}}$.

Evaluation geometry. The operator experiments use the linear kernel $K = XX^\top$ and $A = K + \sigma^2 I$. "Flat-spectrum OOD" means that the covariance eigenvalues are all equal to one random scale $c \sim \text{Unif}[1, 10]$; this spectrum is not present in training. Experiments 1–3 use $n_{\text{ctx}} = 47$ context points. Experiment 1 uses $n_{\text{tgt}} = 3$; Experiments 2–3 use $n_{\text{tgt}} = 16$.

Probe and extraction settings. All finite differences use $\varepsilon = 10^{-3}$ and numerical floor 10^{-12}. Weighted-SVD extraction uses $\tau_{\text{sv}} = 10^{-3}$. For a fixed episode geometry, task-label probes are sampled as $u \sim \mathcal{N}(0, A)$, implemented as $u = z\,\text{chol}(A)^\top$ with $z \sim \mathcal{N}(0, I_{n_{\text{ctx}}})$. Isotropic stress probes are sampled as $u \sim \mathcal{N}(0, I_{n_{\text{ctx}}})$. The interpolation sweep uses independent draws

$$u_\lambda = (1 - \lambda)^{1/2} u_{\text{task}} + \lambda^{1/2} u_{\text{iso}},$$

so its covariance is $(1 - \lambda)A + \lambda I_{n_{\text{ctx}}}$. Experiment 1 uses 4 episodes, 16 build probes, 32 held-out evaluation probes, 4 additivity probe pairs, and $r_{\max} = L + 1 = 9$. Experiment 2 uses 16 episodes per setting, 16 build probes, 16 hidden-validation probes, 8 causal-selection probes, 16 held-out evaluation probes, causal damage threshold 0.05, and at most 16 SVD candidates per layer. It includes depth checkpoints $L \in \{2, 4, 6, 8, 12\}$, head checkpoints $H \in \{1, 2, 8\}$, and input-dimension checkpoints $d_x \in \{3, 5, 8, 10, 15\}$, plus an additional $d_x = 30$ stress checkpoint trained with the same 10k-step recipe. Experiment 3 uses 32 episodes, 32 causal probes, $r_{\max} = 12$, and removes $k \in \{1, 2, 4\}$ directions. Auxiliary rank checks use 1024 sampled geometries per d_x at $(\epsilon, \alpha) = (0.01, 0.05)$. Reported random seeds are 123 for Experiment 1 and 42 for Experiments 2–3.

The setup deliberately uses flat-spectrum evaluation geometries for the operator experiments. This makes the rank requirement easy to interpret and keeps the geometry separate from the spectral curriculum used in training. Task-label probes then test the label directions induced by the sampled evaluation geometry. Experiment 1 also includes isotropic probes as a stricter stress test; Experiments 2–3 report task-label probes because their claim is task-local.

5.2 Experiment 1: Final-State Operator Certificate

Question. Does the transformer locally behave like exact KRR, and if so, can that behavior be explained through a subspace visible in the final context-token activation state? The first test is behavioral: on held-out label perturbations, is $D_\varepsilon F(y)$ close to the exact ridge operator T? Only after that test passes does it make sense to ask whether the agreement factors through an activation-derived reduced operator.

For any frozen basis Q, the operator residual decomposes as

$$D_\varepsilon F(y)[u] - Tu = \big(D_\varepsilon F(y)[u] - T_Q u\big) + (T_Q - T)u.$$

Thus Experiment 1 is organized around three checks. First, the transformer should be close to exact KRR: $E_\varepsilon(F, T)$ should be small. Second, the reduced operator T_Q should be close to exact KRR. Third, the transformer should be close to T_Q. If only the first condition held, we would have behavior without a representation certificate. If only $T_Q \approx T$ held, the subspace would be sufficient for KRR but not tied to the transformer's response. If only $F \approx T_Q$ held, the transformer and the subspace could agree on the wrong operator.

Procedure. For each episode, fix G and compute A, $T = K_t A^{-1}$, and Ty. Draw task-label build probes and held-out evaluation probes. Extract Q_L^{raw} and Q_L^{resp}. Here Q_L^{raw} is the basis extracted from the unperturbed final context-token hidden-state matrix H_L^{ctx}: it uses all context-side directions visible in the final activation state. Q_L^{resp} is extracted instead from finite-difference changes of that final hidden state under task-label build perturbations: it keeps only directions that move when the context labels are changed along directions generated by the task. Both bases are then frozen. We evaluate them on held-out task-label probes and on held-out isotropic probes; the isotropic rows are therefore a transfer test of the task-extracted bases, not a new isotropic extraction. For each basis, form the reduced operator $T_Q = K_t Q Q^\top$.

The comparison control is a same-span response-fitted low-rank operator. It uses the same context-side features $Q^\top u$ as the reduced operator, but it does not use the KRR coefficients $K_t Q$. Instead, it fits a free target-side coefficient matrix to the transformer's finite-difference responses on build probes:

$$C_Q \in \arg\min_{C \in \mathbb{R}^{n_{\text{tgt}} \times r}} \sum_{u \in U_{\text{build}}} \|D_\varepsilon F(y)[u] - C Q^\top u\|_2^2, \qquad T_{\text{actLR}} := C_Q Q^\top.$$

Thus, on a new perturbation u, $T_{\text{actLR}} u$ first computes the same coordinates $Q^\top u$ and then maps those coordinates to target responses using the fitted matrix C_Q. The subscript actLR abbreviates "activation-span low-rank response fit." This control answers a decodability question: using the same subspace coordinates, how well can a freely fitted target-side map reproduce the transformer's local responses? It is not the KRR certificate, because its target-side coefficients are chosen from the transformer rather than from the KRR geometry. Then evaluate T, T_Q, and T_{actLR} on held-out probes. The reported comparison also includes random A-orthonormal subspaces of the same rank; these fail by one to two orders of magnitude on task probes and are used only to exclude the explanation that any subspace of the same dimension would suffice.

Results. The first check passes: the transformer is already close to exact KRR on the task-local perturbations we care about. On the sampled task labels, the transformer has MSE 0.0161 against the noiseless teacher targets, while exact KRR has MSE 0.0148; the ratio is 1.188. The pointwise relative transformer–KRR discrepancy is 0.01045. On held-out task-label perturbations, the finite-difference transformer–KRR error is $E_\varepsilon(F,T) = 0.0121$. This establishes the behavioral target that the rest of the experiment tries to explain through final activations. Additivity defects are much smaller than the operator errors: 4.4×10^{-4} on task probes and 3.5×10^{-3} on isotropic probes at $\varepsilon = 10^{-3}$, with the same scale at 10ε.

Table 1: Behavioral KRR check for the trained transformer, mean over 4 episodes. The operator error is measured on held-out task-label perturbations.

Quantity	Transformer	Exact KRR	Transformer–KRR
Pointwise MSE to teacher targets	0.0161	0.0148	–
Pointwise relative discrepancy	–	–	0.01045
Held-out operator error $E_\varepsilon(F,T)$	–	–	0.0121

We then ask whether this behavioral agreement can be explained through the final activation subspaces.

Table 2: Reduced operator vs. same-span response-fitted low-rank control, mean over 4 episodes. $E(F,S)$ denotes transformer-to-S operator error; $E(S,T)$ denotes S-to-ridge operator error.

Probe distribution	Basis	$E(F,T_Q)$	$E(F,T_{\mathrm{actLR}})$	$E(T_Q,T)$	$E(T_{\mathrm{actLR}},T)$
Task probes	raw	0.01194	0.00726	0.00148	0.01202
Task probes	response	0.01239	0.01169	0.00367	0.01645
Isotropic probes	raw	0.2915	0.2007	0.05254	0.2687
Isotropic probes	response	0.3193	0.2418	0.1278	0.2929

The raw final-state basis gives the cleanest certificate. On task probes, T_Q is very close to exact KRR ($E(T_Q,T) = 0.00148$), and the transformer is close to T_Q ($E(F,T_Q) = 0.01194$). The response-fitted control fits the transformer slightly better (0.00726), but it is much farther from exact KRR (0.01202). This does not make the control a failure; it shows that optimizing the readout for transformer fit is a different question from certifying that the same subspace supports the KRR operator. The response-only basis gives the same task-probe pattern, with T_Q still close to KRR (0.00367) and the response-fitted control farther from KRR (0.01645).

Random subspaces of the same ranks do not explain the result. On task probes, the raw random control has $E(T_Q,T) = 0.117$ and $E(F,T_Q) = 0.118$, roughly two orders of magnitude worse than the raw final-state basis. The response random control is similar, with $E(T_Q,T) = 0.0955$ and $E(F,T_Q) = 0.0955$.

The result is task-local. At the isotropic endpoint, the transformer–KRR operator error rises to about 0.302. For the raw basis, the reduced operator also degrades but remains closer to KRR than the transformer response: $E(T_Q,T) = 0.0525$ while $E(F,T_Q) = 0.292$. The task-built response-only basis also degrades off task: $E(T_Q,T) = 0.128$ and $E(F,T_Q) = 0.319$ on isotropic probes. Thus the response basis is useful as a task-local certificate, but it is not a global explanation of arbitrary label directions. The interpolation sweep in Figure 2 shows this transition from task-label directions to isotropic directions.

Interpretation. Experiment 1 gives the endpoint certificate used by the rest of the note. On task-label perturbations, the transformer is close to KRR, the raw final activation subspace induces a reduced operator close to KRR, and the transformer is close to that reduced operator.

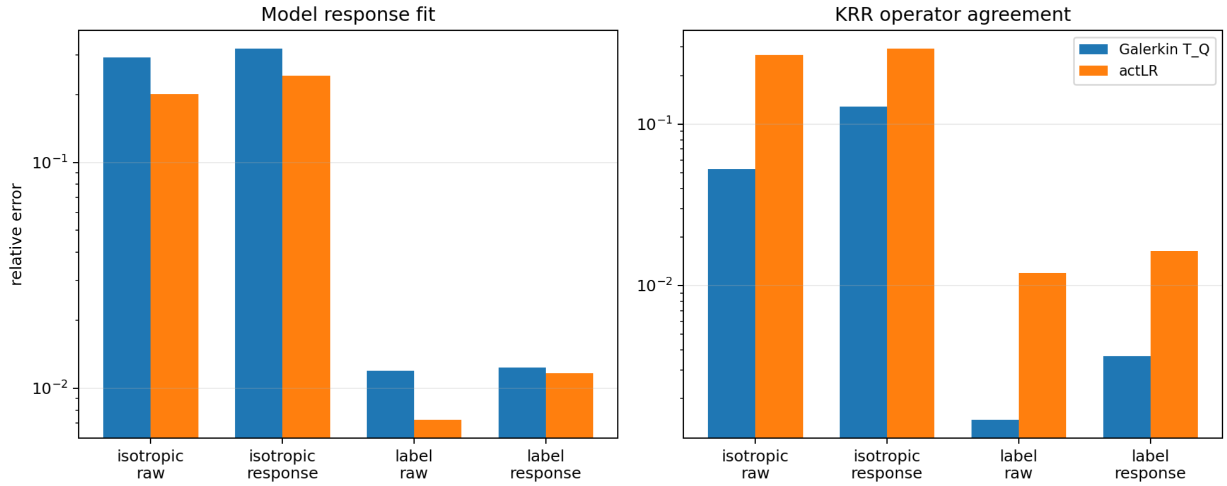

Figure 1: Reduced operator T_Q vs. the same-span response-fitted low-rank control T_{actLR}. On task probes, the response-fitted control fits the transformer slightly better but agrees substantially worse with the ridge operator.

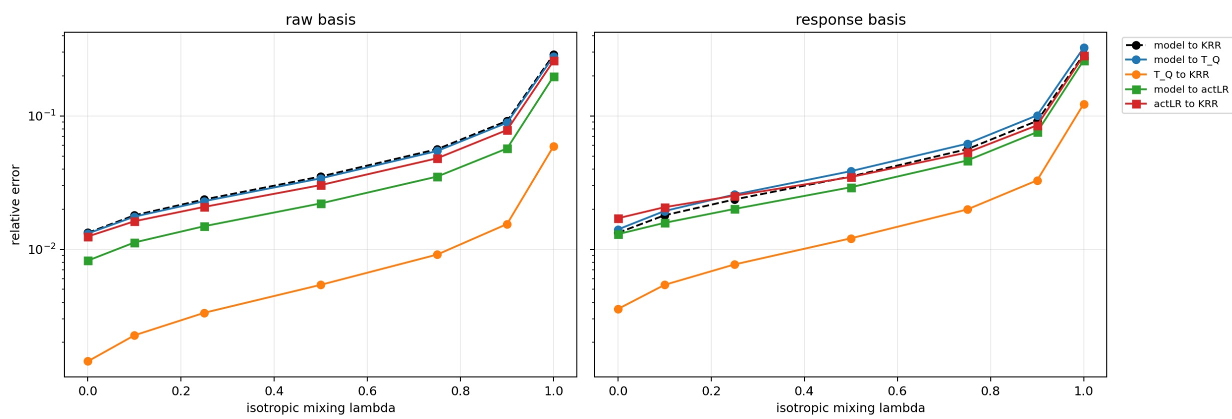

Figure 2: Interpolation sweep from task-label probes to isotropic probes. The interpolation parameter is λ, with $\lambda = 0$ equal to task probes and $\lambda = 1$ equal to isotropic probes. The reduced operator remains comparatively close to ridge while the transformer's learned response drifts off distribution.

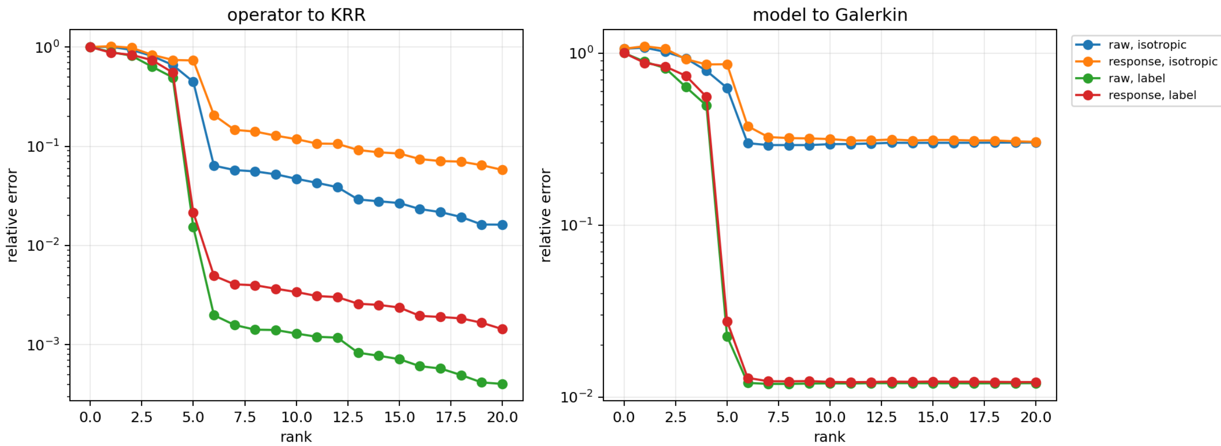

Figure 3: Rank curves for the raw and response-only bases.

The response-fitted control shows that a better fitted readout from the same subspace is not automatically a better KRR explanation: it improves transformer fit slightly but moves away from KRR. The random-subspace controls show that the result is not a generic consequence of using a low-dimensional subspace of the same rank. The isotropic probes show the limit of the claim: Experiment 1 certifies a task-local operator explanation, not a global KRR implementation on all label directions. This experiment is therefore the necessary endpoint check: without it, the layerwise and causal experiments could only show that the transformer contains and uses some low-dimensional activation directions, not that those directions realize the task-local KRR operator we want to explain.

5.3 Experiment 2: Causal Layerwise Rank and Task Rank

Question. Does the layerwise activation span contain the task-visible directions needed for ridge regression? Experiment 1 checked the endpoint: a final activation subspace can induce a reduced operator close to exact KRR. Experiment 2 asks how this subspace is supplied across layers. In the linear task, the required context-label dimension is the task-visible rank r_T, which agrees with d_x in the target-rank sweep. The question is therefore concrete: after we keep only directions that are both visible in layerwise label responses and causally useful for the transformer response, does their accumulated span reach r_T?

Procedure. For each episode, fix the sampled geometry and compute $A = K + \sigma^2 I$ and $T = K_t A^{-1}$. Draw four independent task-label probe sets: build probes, hidden-validation probes, causal-selection probes, and held-out evaluation probes. The build probes propose candidate directions, the causal-selection probes decide which candidates are accepted, and the hidden-validation probes report closure diagnostics. The evaluation probes are used only after the span is frozen.

At layer ℓ, compute hidden-response matrices M_ℓ^{build} and M_ℓ^{val} from finite differences of the context-token hidden state. Suppose earlier layers have already selected directions whose accumulated span before layer ℓ has A-orthonormal basis Q_ℓ^{pre}. Remove the part already explained by this earlier span:

$$N_\ell^{\mathrm{build}} = \left(I - Q_\ell^{\mathrm{pre}}(Q_\ell^{\mathrm{pre}})^\top A\right) M_\ell^{\mathrm{build}}, \qquad N_\ell^{\mathrm{val}} = \left(I - Q_\ell^{\mathrm{pre}}(Q_\ell^{\mathrm{pre}})^\top A\right) M_\ell^{\mathrm{val}}.$$

Thus N_ℓ^{build} is the novel label-sensitive layer response: it is what remains after earlier accepted directions, propagated through the same A-geometry used by the reduced operator, have been accounted for.

Candidate directions are obtained from the A-weighted SVD

$$A^{1/2}N_\ell^{\mathrm{build}} = U_\ell \Sigma_\ell V_\ell^\top, \qquad q_i = A^{-1/2}U_{\ell,i}.$$

The run keeps at most 16 candidates with singular value at least $10^{-3}\sigma_1$. These candidates are not automatically accepted. For each candidate q_i, perform a causal removal after layer ℓ: replace the context hidden state H_ℓ^{ctx} by

$$H_\ell^{\mathrm{ctx}} - q_i q_i^\top A H_\ell^{\mathrm{ctx}},$$

roll the remaining layers forward, and measure the finite-difference response damage on the causal-selection probes:

$$\Delta_\ell(q_i) = \frac{\|D_\varepsilon F_{\ell,q_i}^{\mathrm{rem}}(y) - D_\varepsilon F(y)\|_F}{\|D_\varepsilon F(y)\|_F + \eta_{\mathrm{num}}}.$$

Here $F_{\ell,q_i}^{\mathrm{rem}}$ denotes the transformer after removing direction q_i from the context residual stream at layer ℓ. A candidate is accepted when $\Delta_\ell(q_i) \geq 0.05$. The accepted directions at all layers define Q_{lay} through the layerwise activation-span construction. Finally, freeze Q_{lay}, form $T_{Q_{\mathrm{lay}}} = K_t Q_{\mathrm{lay}} Q_{\mathrm{lay}}^\top$, and evaluate $E(T_{Q_{\mathrm{lay}}}, T)$ and $E_\varepsilon(F, T_{Q_{\mathrm{lay}}})$ on held-out task-label probes.

Results. The 16-episode target-rank sweep uses the causal selection rule described above. Let r_T be the empirical task-visible rank of the ridge operator and let $d_{\mathrm{lay}} = \dim R_{\mathrm{lay}}$. The results are:

Table 3: Experiment 2 target-rank sweep with causal layerwise selection.

d_x	r_T	d_{lay}	d_{lay}/r_T	$E(T_{Q_{\mathrm{lay}}}, T)$	$E(F, T)$	$E(F, T_{Q_{\mathrm{lay}}})$
3	3.0	3.0	1.000	0.0000	0.0063	0.0063
5	5.0	5.0	1.000	0.0000	0.0091	0.0091
8	8.0	8.0	1.000	0.0000	0.0129	0.0129
10	10.0	10.0	1.000	0.0000	0.0131	0.0131
15	14.9	14.5	0.971	0.0997	0.0174	0.1089
30	16.0	21.4	1.336	0.5402	1.6873	1.5633

For $d_x = 3, 5, 8, 10$, the selected layerwise span reaches the task-visible rank and the reduced operator is numerically exact on held-out task probes. For $d_x = 15$, the selected dimension averages 14.5 while r_T averages 14.9, and the reduced-operator error rises to 0.0997. The threshold is therefore visible in the held-out operator error: when the causal layerwise span reaches r_T, $T_{Q_{\mathrm{lay}}}$ matches T; when it remains slightly short, the reduced operator loses accuracy. The held-out closure diagnostics follow the same pattern. The mean maximum validation closure defect is about 0.027, 0.027, 0.026, 0.015 for $d_x = 3, 5, 8, 10$, and rises to 0.110 for $d_x = 15$.

The $d_x = 15$ case appears to fail for a simple reason: the selected span does not reliably recover the last task-visible direction. This is the largest target-rank setting, so the required subspace is nearly the full 15-dimensional feature span. The causal extraction often selects 14 rather than 15 independent directions, and the validation closure defect is higher, meaning that the accepted span no longer covers all held-out layer responses. In the episodes where the selected span reaches the full task-visible rank, the reduced-operator error is again numerical zero; the nonzero average error comes from the episodes that miss one direction. Thus the failure is best read as incomplete rank recovery by the layerwise extraction rule at the largest tested dimension, not as evidence that the reduced-operator construction changes.

The additional $d_x = 30$ row is a stress check rather than a clean continuation of the earlier rank sweep. Because this experiment uses $n_{\mathrm{tgt}} = 16$, the task-visible rank is capped at 16 even though the input dimension is 30. The $d_x = 30$ checkpoint was trained with the same architecture and 10k-step recipe as the other d_x-sweep models, but its pointwise training evaluation remains far from ridge. Correspondingly, the causal layerwise extraction accepts many directions, $d_{\mathrm{lay}} = 21.4$, yet the induced reduced operator is not close to ridge:

$$E(T_{Q_{\mathrm{lay}}}, T) = 0.5402, \qquad E(F, T) = 1.6873.$$

Thus $d_x = 30$ should be read as a high-dimensional undertrained/capacity stress case: having more accepted directions than the capped task-visible rank does not help if the selected span is not aligned with the task-visible KRR directions and the transformer itself is not in the KRR-like behavioral regime.

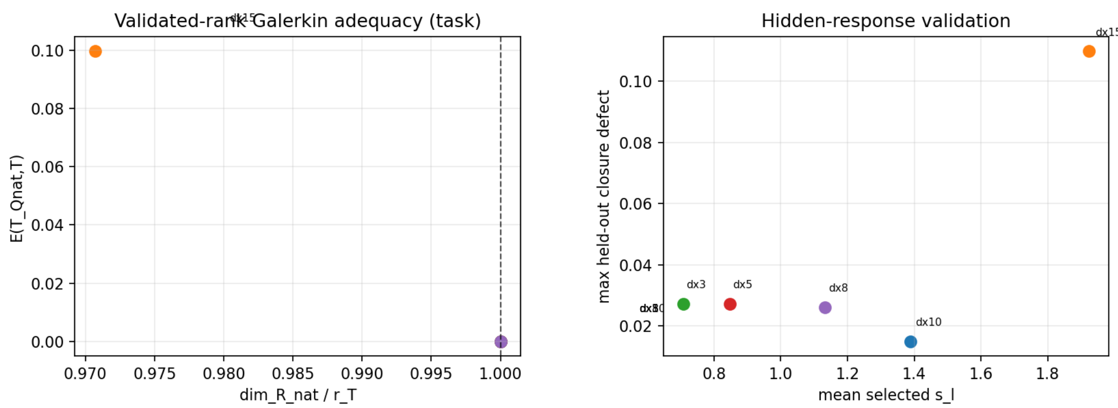

Figure 4: Experiment 2 target-rank sweep with causal layerwise selection for the original $d_x \leq 15$ sweep. When the selected layerwise dimension reaches the task-visible rank, the reduced-operator error collapses to numerical zero. The $d_x = 15$ setting remains slightly under-ranked.

Table 4: Experiment 2 fixed-$d_x = 5$ depth/head sweep with causal layerwise selection, mean over 16 episodes per checkpoint. All checkpoints use width $D = 128$. B_{nat} is the pre-dependency native count, d_{lay} is the accepted span dimension, and valClos is the maximum validation closure defect.

Checkpoint	L	H	B_{nat}	d_{lay}	r_T	d_{lay}/r_T	mean s_ℓ	$E(T_Q, T)$	$E(F, T_Q)$	$E(F, T)$	valClos
$H = 1$	8	1	60.6	5.0	5.0	1.000	0.90	0.00000	0.01560	0.01560	0.04822
$H = 2$	8	2	61.2	5.0	5.0	1.000	0.98	0.00000	0.00904	0.00904	0.03302
$L = 2$	2	4	37.5	20.6	5.0	4.112	5.42	0.00000	0.18535	0.18535	0.01672
$L = 4$	4	4	29.2	5.1	5.0	1.025	1.21	0.00000	0.02528	0.02528	0.01555
$L = 6$	6	4	41.6	5.0	5.0	1.000	0.88	0.00000	0.01255	0.01255	0.02851
$H = 8$	8	8	57.2	5.0	5.0	1.000	0.79	0.00000	0.00982	0.00982	0.03199
$L = 8$	8	4	58.3	5.0	5.0	1.000	0.83	0.00000	0.00952	0.00952	0.03626
$L = 12$	12	4	109.8	5.0	5.0	1.000	0.93	0.00000	0.00944	0.00944	0.02912

The depth/head sweep gives the necessary qualification. This is a fixed-d_x sweep: all these checkpoints are trained and evaluated at $d_x = 5$, with width $D = 128$, while depth varies over

$L \in \{2,4,6,8,12\}$ at four heads and head count varies over $H \in \{1,2,4,8\}$ at eight layers. Causal selection gives $E(T_{Q_{\text{lay}}}, T) = 0$ for all reported checkpoints, but transformer accuracy still varies. The $L = 2$ checkpoint has $d_{\text{lay}} = 20.6$ and exact reduced-operator error, yet $E(F,T) = 0.185$. By contrast, the $L = 8$ checkpoint has $d_{\text{lay}} = 5.0$, exact reduced-operator error, and $E(F,T) = 0.0095$; the $H = 2$ and $L = 12$ checkpoints are similar, with transformer errors 0.0090 and 0.0094. The fixed-d_x sweep therefore separates two facts: the activations can expose a subspace from which KRR is recoverable, while the trained transformer may still fail to use that available subspace accurately. The layerwise span certifies availability of a ridge-sufficient operator subspace, not automatic behavioral KRR agreement.

Interpretation. Experiment 2 supports a sharper resource law:

$$d_{\text{lay}}/r_T \approx 1 \quad \Longrightarrow \quad E_{P_{\text{task}}}(T_{Q_{\text{lay}}}, T) \approx 0.$$

The law concerns the reduced operator induced by the causal layerwise span. It is not a statement about raw architecture counts, and it is not by itself a statement about transformer accuracy. This experiment was needed because Experiment 1 only showed that a suitable subspace is present at the final state; it did not show whether the layerwise label responses supply the task-visible number of directions. Experiment 2 shows that, for the linear task, the causally selected layerwise span reaches exactly the dimension predicted by the KRR geometry in the successful target-rank settings, and that operator error appears precisely when this span remains short.

5.4 Experiment 3: Causal Direction Removal

Question. Are the task-important directions of the reduced operator causally used by the transformer? This is needed because a direction can be decodable without being used. The surgery test removes directions ranked important by the reduced operator and checks whether the transformer's task-local response changes accordingly.

Procedure. For an extracted raw final A-orthonormal basis $Q = [q_1, \dots, q_r]$, define Q_{-j} by removing column j from Q. Direction q_j is important for the reduced operator if deleting it changes the reduced operator's response. On a probe set U, measure this by

$$\Lambda_j^{\text{red}}(U) := \left(\frac{\sum_{u \in U} \|(T_Q - T_{Q_{-j}})u\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

Task leverage means this score computed on task-label probes; isotropic leverage means the same score computed on isotropic probes. The main intervention set consists of the top-k directions by task leverage.

We intervene on the context-token residual state after residual state $s = 0, \dots, L$ by replacing

$$H_s^{\text{ctx}} \longmapsto H_s^{\text{ctx}} - \sum_{j \in J} q_j q_j^\top A H_s^{\text{ctx}},$$

then roll the remaining layers forward. The intervention removes the A-orthogonal projection of the context residual stream onto directions that the reduced operator predicts to be important. Matched controls remove low-task-leverage directions from the same extracted basis, high-isotropic-but-low-task directions, high-activation-variance-but-low-task directions, and random subsets of Q. The low-task directions are the cleanest negative control because they share the same extraction pipeline and residual-stream coordinates.

Results. Before surgery the checkpoint is in the task-local reduced-operator regime:

$$E_{\text{task}}(F,T) = 0.00926, \qquad E_{\text{task}}(F,T_Q) = 0.00920, \qquad E_{\text{task}}(T_Q,T) = 0.00086.$$

The same episodes retain the off-distribution response drift seen above:

$$E_{\text{iso}}(F,T) = 0.266, \qquad E_{\text{iso}}(T_Q,T) = 0.0378.$$

Surgery is therefore applied in a regime where the task-local certificate holds before intervention. At residual state $s = 1$, removing a single direction predicted to matter most for the reduced operator causes prediction drift 4.68, KRR-error inflation 649×, and task response damage 41.59. Removing a single low-task-leverage direction from the same basis is nearly inert: prediction drift $9.8 \cdot 10^{-4}$, KRR-error inflation 1.03×, and task response damage 0.0024.

The layer sweep localizes the effect. High-task surgery is most damaging at early residual states, before later attention has routed the relevant context information into target tokens. Context-only surgery after the final layer is a no-op, as expected from the architecture. At $s = 1$, the relationship between aggregate task leverage and task response damage is strong across matched non-random controls (log–log Pearson 0.91, Spearman 0.87). This is the key causal specificity check: damage tracks task leverage, not merely activation energy or membership in the extracted subspace.

As a complementary mediation check, we also intervene on the whole extracted subspace at residual state $s = 1$. Keeping only the A-projection of the context residual stream onto Q,

$$H_s^{\text{ctx}} \mapsto QQ^\top A H_s^{\text{ctx}},$$

nearly preserves the task-local response: over 8 episodes,

$$E_{\text{task}}(F,T) = 0.01866, \qquad \text{response damage} = 0.01659.$$

Removing the same subspace,

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - QQ^\top A H_s^{\text{ctx}},$$

destroys the response:

$$E_{\text{task}}(F,T) = 145.43, \qquad \text{response damage} = 145.43.$$

Thus, in this intervention family, the complement of Q is largely dispensable for the task-local KRR response, while Q itself is necessary.

Interpretation. The directions that the reduced operator identifies as task-relevant are not only decodable and sufficient for small reduced-operator error; removing them before later attention can use them strongly damages both prediction and the task-local finite-difference response.

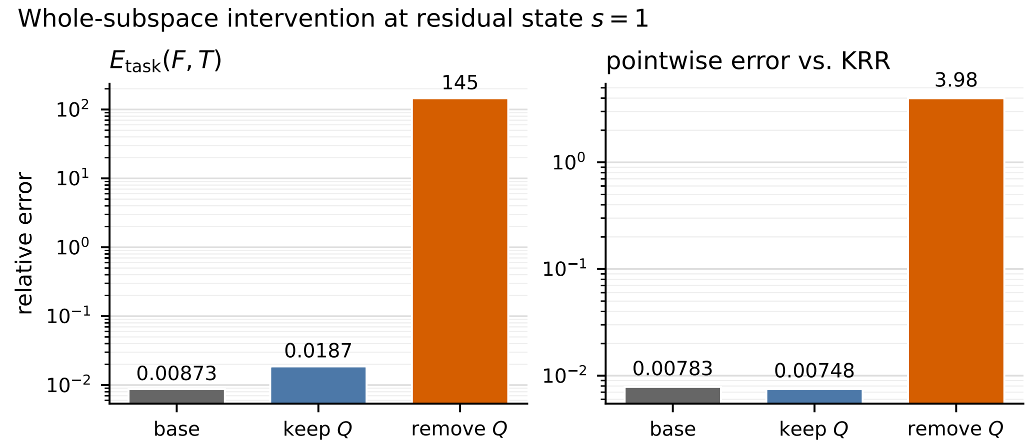

Figure 5: Experiment 3: whole-subspace complement ablation at residual state $s = 1$. Keeping only the A-projection onto Q nearly preserves both the local task response and the pointwise KRR prediction, while removing the same subspace destroys both.

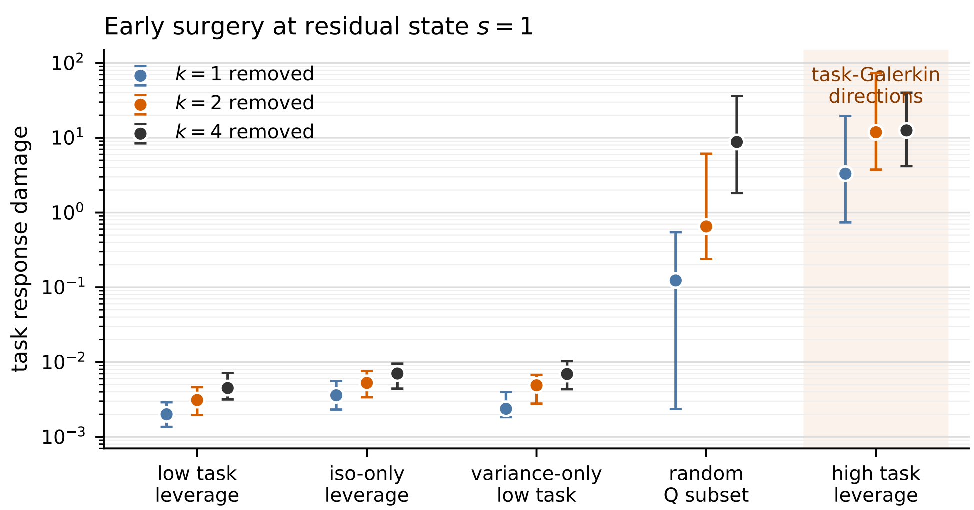

Figure 6: Experiment 3: causal effect of task-important activation directions. Task finite-difference response damage after removing k context-token directions at residual state $s = 1$. High task-leverage directions produce orders-of-magnitude larger damage than low-task-leverage directions from the same activation basis.

6 Falsification and Conclusion

The expected observation would be only that a transformer trained on linear regression gives KRR-like predictions. That is not the main claim here. Pointwise prediction accuracy alone does not identify the computation inside an in-context learner: many rules can agree on the sampled labels while having a different response to nearby label perturbations.

The stronger claim is operator-level and task-local. After fixing the input geometry, KRR is the linear map $T = K_t A^{-1}$ from context labels to target predictions. We therefore ask whether the transformer has the same local label-to-prediction response, whether that response can be explained through a compact activation subspace, and whether the relevant directions in that subspace are causally used.

The evidence has three parts. First, on held-out task-label perturbations, the transformer's finite-difference response is close to the exact KRR operator. Second, the activation-derived subspace becomes sufficient exactly when its dimension reaches the task-visible rank predicted from the KRR spectrum. This rank is computed before looking at activations; it is the number of directions KRR needs under the task label distribution. Third, the high-leverage directions of the reduced operator are not merely decodable: removing them from the residual stream strongly damages prediction and the task-local response, while matched low-leverage removals are nearly inert.

These observations make the result more specific than a generic low-rank explanation. The same-span response-fitted low-rank control can fit the transformer's response slightly better on task probes, but it agrees much worse with KRR. The reduced operator used here is constrained by the KRR system itself, so success means that the extracted subspace supports the ridge computation, not just that some low-rank response fit works after the fact.

The claim is also not global. Isotropic label probes show clear response drift: the transformer has not learned a uniformly correct inverse on all possible label directions. The supported conclusion is narrower and more mechanistic. On the label directions emphasized by the training distribution, the trained transformer exposes a compact context-side activation subspace whose induced KRR operator is accurate, whose required dimension is predicted by the task-visible rank, and whose high-leverage directions are causally important for the transformer's response.

The account is falsifiable at each step. If $E_\varepsilon(F, T)$ is large while pointwise error is small, the transformer has not learned the conditional KRR response. If $E(T_Q, T)$ is large, the extracted subspace does not explain the KRR operator. If $E(T_Q, T)$ is small but $E_\varepsilon(F, T_Q)$ is large, the subspace is sufficient but the transformer does not use it to produce its local response. If T_{actLR} explains held-out responses much better than T_Q while also staying close to KRR, the reduced-operator construction is not the right explanation. If leverage does not predict causal damage, the directions may be decodable but not causally operative.